

HOBOKSAR, China

WMO: 511560

Lat: 46.8103N

Lon: 85.7498E

Elev: 1323

StdP: 86.41

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-21.0	-28.9	0.3	-21.7	-26.8	0.4	-19.1	10.4	-9.9	8.1	-9.3	1.4	320	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.5	14.8	27.9	14.2	26.4	13.7	16.2	25.1	15.6	24.4	14.9	23.5	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.3	11.2	19.0	12.3	10.5	18.8	11.3	9.8	18.8	50.5	24.9	48.3	24.5	46.1	23.6	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	8.4	6.7	-25.4	32.1	2.6	1.9	-27.3	33.5	-28.8	34.6	-30.3	35.7	-32.2	37.1		
			-25.7	17.4	2.6	1.2	-27.5	18.3	-29.0	19.0	-30.5	19.7	-32.4	20.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-12.3	-9.1	-1.7	6.9	12.9	18.3	20.0	18.5	12.6	5.0	-3.5	-10.0	(6)
(7)		DBStd	12.20	4.36	4.76	5.71	5.04	4.33	3.36	2.95	3.24	4.35	4.83	5.33	4.27	(7)
(8)		HDD10.0	2943	692	536	363	114	22	1	0	1	23	165	406	621	(8)
(9)		HDD18.3	5081	951	769	621	343	175	41	16	38	178	414	656	879	(9)
(10)		CDD10.0	1064	0	0	0	21	111	250	309	263	101	9	0	0	(10)
(11)		CDD18.3	159	0	0	0	0	5	40	67	42	5	0	0	0	(11)
(12)		CDH23.3	1303	0	0	0	3	56	309	537	353	46	0	0	0	(12)
(13)		CDH26.7	288	0	0	0	0	6	59	142	75	7	0	0	0	(13)
(14)	Wind	WSAvg	2.3	1.4	1.6	2.2	3.1	3.3	2.9	2.7	2.6	2.2	1.8	1.5	(14)	
(15)	Precipitation	PrecAvg	136	3	2	5	7	12	28	37	19	11	5	4	3	(15)
(16)		PrecMax	209	12	10	20	18	31	75	73	50	32	16	17	13	(16)
(17)		PrecMin	70	0	0	0	0	0	4	7	4	1	0	0	0	(17)
(18)		PrecStd	37	3	2	4	6	8	15	20	11	9	4	4	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	3.8	15.0	22.7	27.2	30.2	32.5	30.9	27.6	20.0	11.2	1.8	(19)
(20)			MCWB	-1.9	-0.8	5.5	10.0	12.4	14.7	16.0	15.1	13.1	8.8	5.0	-1.5	(20)
(21)		2%	DB	-2.4	1.2	11.8	19.4	24.9	28.4	30.0	28.7	24.4	17.0	8.7	-0.5	(21)
(22)			MCWB	-4.7	-2.3	4.2	8.3	11.5	14.1	15.4	14.3	11.8	7.3	3.6	-3.3	(22)
(23)		5%	DB	-4.5	-0.5	9.3	17.1	22.8	26.8	28.2	27.0	22.2	14.7	6.2	-2.5	(23)
(24)			MCWB	-6.5	-3.6	3.1	7.0	10.7	13.6	14.6	13.9	10.9	6.5	1.9	-4.7	(24)
(25)		10%	DB	-6.2	-2.3	6.8	15.0	20.7	25.0	26.4	25.2	20.2	12.4	3.6	-4.2	(25)
(26)			MCWB	-8.0	-5.0	1.7	6.4	9.7	13.1	14.2	13.5	10.1	5.6	0.0	-6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-0.1	6.5	11.1	13.8	16.1	18.0	17.1	13.7	9.6	6.0	-1.2	(27)
(28)			MCDWB	0.7	2.6	13.5	21.7	24.2	25.6	27.9	25.3	24.1	17.9	9.9	1.5	(28)
(29)		2%	WB	-4.4	-2.0	4.8	9.3	12.5	15.4	16.7	15.8	12.7	8.2	3.8	-3.0	(29)
(30)			MCDWB	-2.7	0.8	10.9	17.3	22.0	24.7	25.3	25.3	22.3	15.1	8.1	-1.0	(30)
(31)		5%	WB	-6.4	-3.5	3.5	8.1	11.6	14.7	15.9	15.1	11.7	7.1	1.9	-4.6	(31)
(32)			MCDWB	-4.7	-0.8	8.8	15.1	20.8	23.8	24.3	23.9	20.5	13.6	6.0	-2.6	(32)
(33)		10%	WB	-7.9	-4.9	2.0	6.9	10.6	14.0	15.2	14.3	10.7	6.0	0.3	-6.0	(33)
(34)			MCDWB	-6.2	-2.5	6.5	13.8	19.1	23.0	23.7	23.0	19.2	11.8	3.3	-4.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.4	9.7	10.5	10.6	10.0	9.9	10.7	10.8	9.8	7.9	8.0	(35)
(36)			MCDBR	9.3	10.3	11.7	13.0	12.7	11.9	12.2	12.6	12.9	12.4	10.2	9.2	(36)
(37)			MCWBR	7.5	7.7	6.6	6.4	5.5	4.5	4.1	4.8	5.7	6.6	6.5	7.2	(37)
(38)		5% WB	MCDBR	9.0	9.8	10.9	10.7	10.9	10.2	9.8	10.6	11.4	11.2	9.8	8.8	(38)
(39)			MCWBR	7.3	7.4	6.5	5.7	5.0	4.1	3.8	4.4	5.4	6.2	6.6	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.298	0.339	0.386	0.374	0.387	0.386	0.363	0.334	0.321	0.292	0.270	(40)	
(41)		taud	2.211	2.153	2.150	2.136	2.243	2.272	2.307	2.353	2.387	2.330	2.301	2.219	(41)	
(42)		Ebn at Noon	815	860	875	865	889	877	873	881	878	828	781	763	(42)	
(43)		Edh at Noon	82	108	127	143	133	130	124	114	100	90	75	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.79	2.83	4.07	5.32	6.59	6.82	6.55	5.98	4.78	2.97	1.67	1.30	(44)	
(45)		RadStd	0.14	0.15	0.29	0.30	0.35	0.35	0.43	0.28	0.27	0.24	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page